

From Simulation to the Real-World: An In-Field 6D Pose Dataset and Baseline for Robotic Strawberry Harvesting

Woojung Son*, Won Suk Lee*, Zijing Huang*, Daeun Choi*, Catia Silva[†], Yu She[‡], and Yan Gu[§]

*Department of Agricultural and Biological Engineering, University of Florida
{w.son, wslee, zijing.huang, dana.choi}@ufl.edu

[†]Department of Electrical and Computer Engineering, University of Florida, catiaspsilva@ece.ufl.edu

[‡]Edwardson School of Industrial Engineering, Purdue University, yushe@purdue.edu

[§]School of Mechanical Engineering, Purdue University, yangu@purdue.edu

Abstract—Robotic strawberry harvesting requires precise 6D pose estimation; however, collecting 6D pose ground truth in real agricultural fields is inherently challenging. Existing strawberry 6D pose estimation studies have therefore relied mainly on synthetic data, leaving their in-field performance unquantified. In this work, we obtain ground truth indirectly, by recovering camera poses via PnP, reconstructing each scene at metric scale, and annotating a single 3D bounding box per strawberry that is propagated across all frames, yielding, to the best of our knowledge, the first real-world 6D pose ground truth dataset of red-stage strawberries collected at an actual strawberry farm (12,040 images). We also introduce a synthetic dataset rendered in NVIDIA Isaac Sim, featuring scene-level realism and domain randomization. Despite this improved simulation setup, models trained on synthetic data alone fail to transfer to in-field images, while introducing a small amount of real data improves both translation and rotation accuracy across all backbones. Under the monocular RGB setting evaluated here, rotation is largely recovered once real data is used, and depth is what limits pose accuracy. These baselines across backbone encoders serve as a reference for future work. The real-world dataset is publicly available at <https://huggingface.co/datasets/WoojungSon/FieldStraw6D>, and the data-collection pipeline is available at <https://github.com/wjson2435/FieldStraw6D-pipeline>.

Index Terms—6D Pose Estimation, Robotic Harvesting, Sim-to-Real Gap

I. INTRODUCTION

Strawberries are among the most widely consumed fruits in the world, but they are highly labor-intensive to produce, with harvesting labor accounting for roughly 40% of total production costs [1], and the industry faces a growing labor shortage. Fully autonomous strawberry harvesting at commercial scale has yet to be realized [2]. Among the challenges, damage-free handling of soft, fragile strawberries requires precise robotic grasping, which depends on knowing not only fruit position but also orientation, that is, the fruit’s full 6D pose. Existing harvesting systems, however, estimate 3D fruit position only [3]–[5].

Despite this need, the few methods that do pursue strawberry orientation or 6D pose estimation [6]–[8] are each evaluated on a different simulator, leaving them mutually

incomparable. On real images they can only be inspected qualitatively, because no means exists of obtaining 6D pose ground truth in the field. Reported accuracy therefore reflects only the training distribution and cannot predict real-world deployment performance, leaving the sim-to-real gap for this task unmeasured. Most consequentially, resolving this gap would make it possible to determine whether continued investment in simulation realism is worthwhile or whether other strategies are needed instead.

Such a means has not existed for two reasons. Physically, field-grown plants cannot be instrumented or marked, and controlled acquisition setups such as turntables are unavailable in the field. Separately, conventional 6D pose annotation pipelines require an exact CAD model per object, an assumption strawberries violate given their variable size, shape, and calyx geometry. Ground truth for agricultural crops has therefore been obtained either in simulation [7], [8], where these obstacles do not arise, or in controlled laboratory settings [9], [10], where the plant can be instrumented and modeled. Each setting removes one obstacle by removing the field itself, which is why both remain unresolved wherever the fruit actually grows.

Because direct in-field measurement is difficult, we obtain ground truth indirectly. We collect video sequences of strawberry plants in real fields, recover camera poses via Perspective-n-Point (PnP) estimation, reconstruct the scene at metric scale, and annotate a 3D bounding box once per instance in the reconstructed space, propagating it across all frames through the recovered camera poses. This yields 12,040 annotated images. In parallel, we render a synthetic dataset in NVIDIA Isaac Sim with scene-level realism and domain randomization, as a control: by improving simulation fidelity over prior work, we narrow, though do not eliminate, the extent to which any residual sim-to-real gap can be explained by inadequate rendering. Building on both datasets, we evaluate three backbone encoders for monocular RGB-only strawberry 6D pose estimation. Models trained purely on synthetic data transfer poorly to the field, while adding even a small amount

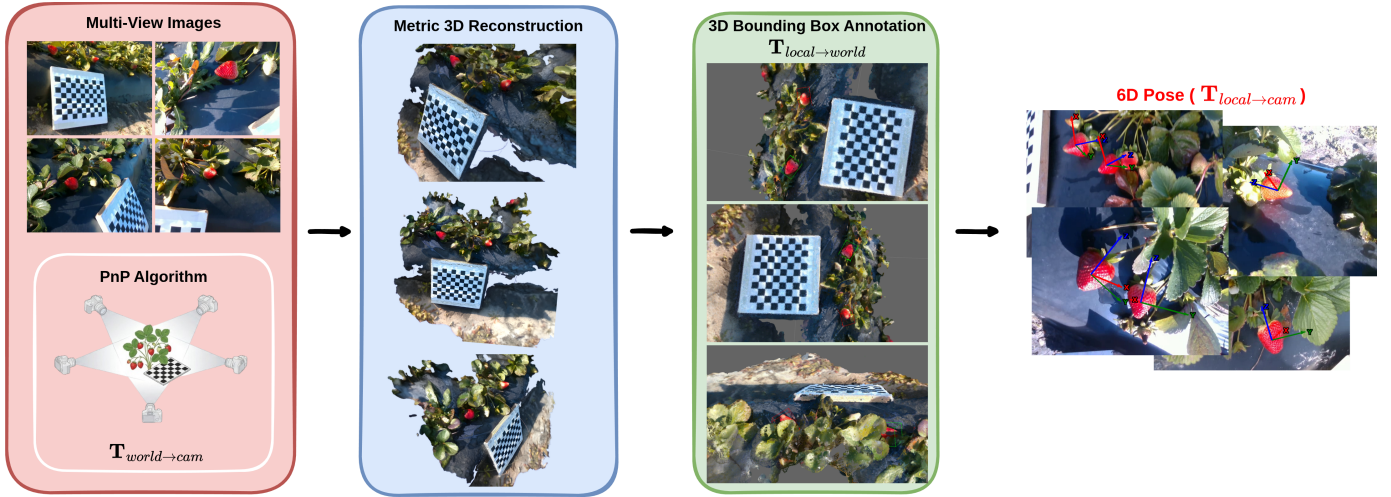


Fig. 1. Overview of the real-world dataset construction pipeline. A video sequence is recorded with a checkerboard placed near the strawberry plant, and PnP is applied to estimate the world-to-camera transformation $\mathbf{T}_{world \rightarrow cam}$ for each frame. Metric 3D reconstruction is performed using COLMAP, and 3D bounding boxes are manually annotated on the resulting point cloud to obtain $\mathbf{T}_{local \rightarrow world}$. In parallel, 2D bounding boxes are manually annotated on the RGB frames. The 6D pose ground truth is finally composed as $\mathbf{T}_{local \rightarrow cam} = \mathbf{T}_{world \rightarrow cam} \cdot \mathbf{T}_{local \rightarrow world}$.

of real data improves both translation and rotation accuracy across all backbones. Under this monocular RGB setting, rotation is largely recovered once real data is used, and depth is what limits pose accuracy.

The main contributions of this work are as follows:

- We present, to the best of our knowledge, the first in-field red-stage strawberry 6D pose ground-truth dataset, containing 12,040 images and 16,037 annotated strawberry instances.
- We introduce a synthetic dataset rendered in NVIDIA Isaac Sim with scene-level realism and domain randomization, serving as a control that narrows the extent to which the measured gap can be attributed to rendering fidelity alone.
- We provide baseline results across three backbone encoders under synthetic-only, mixed, and real-only training, quantifying the sim-to-real gap for strawberry 6D pose estimation in the field.

II. RELATED WORK

Prior robotic strawberry harvesting systems achieve fruit detection and localization using RGB-D 3D centroid estimation for pick-and-place operations [3]–[5], without recovering fruit orientation. Wagner et al. [6] estimate strawberry orientation but not full 6D pose, and Li and Kasaei [7] and Sinha et al. [8] extend this to full 6D pose estimation. All three train purely in simulation and report quantitative accuracy only on synthetic test data; on real images they are restricted to qualitative inspection, as no in-field 6D pose ground truth exists to score against.

While substantial effort has gone into agricultural perception datasets, few provide 6D pose ground truth for agricultural produce. Abdulsalam et al. [9] and Chatzis et al. [10] introduce multi-category produce datasets with 6D pose ground truth, collected in controlled laboratory settings, including through

a calibrated robotic manipulator [10]; neither includes strawberries. Chatzis et al. further note that instance-level 6D pose approaches require exact 3D CAD models, a limitation for produce with high intra-class shape variability.

Synthetic data is widely used in agricultural robotics to reduce annotation cost, but a domain gap between simulated and real environments persists even for simpler tasks such as object detection: Hutter-Mironovová [11] reports that fruit detectors trained exclusively on synthetic images fall well short of real-trained counterparts, and that while adding real images recovers most of this gap in-domain, hybrid models still degrade once the background changes. Because 6D pose estimation demands finer-grained geometric understanding than detection, this gap may be more severe for pose estimation, yet for agricultural produce it remains unmeasured in the field.

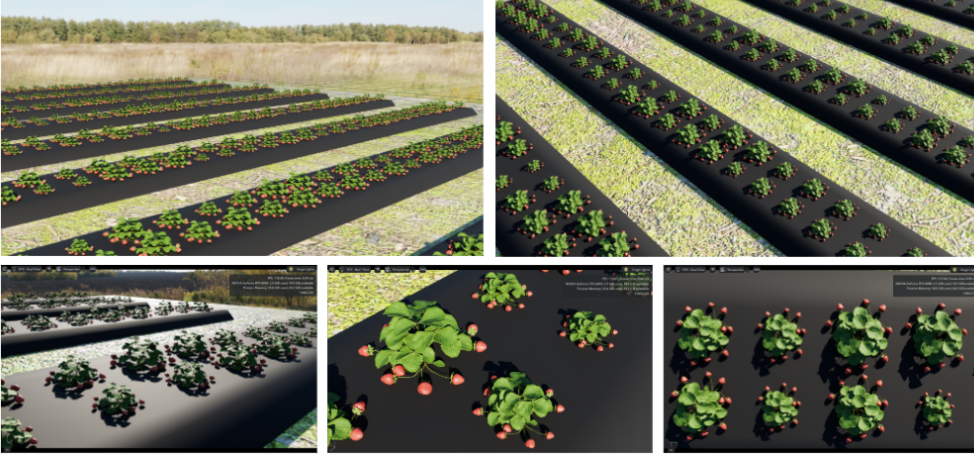
III. METHOD

We first describe the construction of the real-world and synthetic datasets (Sec. III-A), then the baseline 6D pose estimation architecture trained on them (Sec. III-B).

A. Dataset

We collect two complementary datasets for strawberry 6D pose estimation: a real-world dataset of 12,040 images captured in an actual agricultural field, and a synthetic dataset rendered in NVIDIA Isaac Sim. For both datasets, the strawberry’s local coordinate frame is defined with the $+z$ axis pointing toward the stem. The x - and y -axes are not anchored to a specific anatomical feature; they follow the orientation of the annotated (real-world) or modeled (synthetic) bounding box, since strawberries are approximately rotationally symmetric about the stem axis and lack a consistent visual landmark for a canonical azimuthal reference. We revisit this in Sec. IV-B with a symmetry-aware stem-axis error that

(a)



(b)



(c)

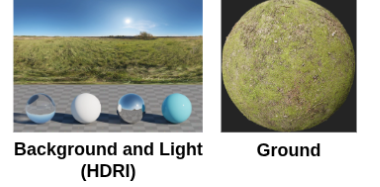


Fig. 2. Synthetic environment in NVIDIA Isaac Sim. (a) Rendered strawberry farm scenes. (b) Strawberry plant models with geometry variation. (c) Background and lighting (HDRI) and ground material.

TABLE I

OVERVIEW OF THE REAL-WORLD AND SYNTHETIC DATASETS. INSTANCES DENOTE THE TOTAL NUMBER OF ANNOTATED STRAWBERRIES ACROSS ALL IMAGES.

	Real-World	Synthetic
Images	12,040	35,118
Instances	16,037	127,910
Resolution	640 × 480 pixels	
Sensor / Renderer	RealSense D435i	Isaac Sim
Environment	Strawberry Farm	Simulated Strawberry Farm
Annotation	6D pose + 2D bbox	

is invariant to this azimuthal ambiguity. Dataset details are summarized in Table I.

Real-World Dataset. The construction pipeline is illustrated in Fig. 1. An Intel RealSense D435i camera is used for all data collection, capturing frames at 640 × 480 pixels resolution. Since robotic harvesting targets only ripe fruit, data collection was restricted to red-stage strawberries. We first calibrate the camera to obtain the intrinsic matrix $\mathbf{K}$, required by the PnP algorithm in the subsequent step.

A checkerboard (11 × 8 squares, A4, 25 mm square size) is placed near the strawberry plant, and a video sequence is recorded from varying distances, within the limits imposed by the surrounding plants and bed layout, to cover viewpoints relevant to robotic manipulation, increase the diversity of 6D pose annotations, and improve the success rate of 3D reconstruction. Frames are extracted from the recorded video, and for frames in which the checkerboard is visible, we apply the PnP algorithm [12] to compute the world-to-camera transformation $\mathbf{T}_{world \rightarrow cam}$. We represent all transformations as 4 × 4 homogeneous matrices:

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \in SE(3), \quad \mathbf{R} \in SO(3), \quad \mathbf{t} \in \mathbb{R}^3 \quad (1)$$

We perform sparse reconstruction using COLMAP [13], [14] on all extracted frames, aligning the reconstructed

model to the world coordinate system using the per-frame $\mathbf{T}_{world \rightarrow cam}$ as reference poses to recover metric scale. Because COLMAP registers all frames of a sequence into a single reconstruction, the checkerboard need only be visible in a subset of frames: those frames anchor the reconstruction to the world coordinate system, and the multi-view constraints between cameras then place every remaining frame in the same coordinate system, yielding $\mathbf{T}_{world \rightarrow cam}$ for all frames.

Dense reconstruction is then performed via PatchMatch stereo and stereo fusion to produce a metric point cloud. A sequence is discarded only when COLMAP fails to produce a reconstruction; no reprojection-error threshold is applied. We report the reprojection errors of the retained sequences as a measure of reconstruction quality in Sec. IV-D.

We manually annotate a 3D bounding box on the reconstructed metric point cloud for each visible strawberry instance, specifying its position, orientation, and dimensions. Each annotated box provides the object’s position and orientation in the world coordinate system as $\mathbf{T}_{local \rightarrow world}$.

The 6D pose of each object with respect to the camera is then obtained by:

$$\mathbf{T}_{local \rightarrow cam} = \mathbf{T}_{world \rightarrow cam} \cdot \mathbf{T}_{local \rightarrow world} \quad (2)$$

To align with the OpenGL coordinate convention adopted by NVIDIA Isaac Sim, the resulting poses are converted from OpenCV by applying a 180° rotation about the x -axis:

$$\mathbf{T}_{OpenGL} = \mathbf{M} \cdot \mathbf{T}_{OpenCV}, \quad \mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

Finally, 2D bounding boxes are manually annotated for instances belonging to successfully reconstructed sequences, i.e., sequences not discarded due to COLMAP reconstruction failure (Sec. IV-D). The resulting real-world dataset comprises

12,040 images and 16,037 annotated strawberry instances, each with a 6D pose and a 2D bounding box.

Synthetic Dataset. Prior synthetic strawberry datasets model plants inside a general-purpose robotics simulator [7] or place fruit among non-agricultural distractor objects [8], without reproducing the surrounding field environment. Our dataset instead models a full strawberry farm scene, with realistic High Dynamic Range Image (HDRI) environment lighting, a textured ground material, and plant-level geometry variation. We construct a strawberry farm environment in NVIDIA Isaac Sim using a red-stage strawberry plant model developed for this study, a CC0-licensed ground material from AmbientCG [15], and a farm field HDRI from Poly Haven [16]. To ensure domain alignment, the synthetic camera is configured to match the Intel RealSense D435i used in real-world collection, including identical image resolution (640×480 pixels) and camera intrinsics.

The camera is positioned on a hemisphere centered on the strawberry plant, with distance sampled uniformly in $[0.2, 1.0]$ m and the viewing direction constrained to face the plant at all times. To reduce positional bias in the synthetic images, a random offset is applied to the look-at target point, so that the camera points toward a location near, but not exactly at, the strawberry center, causing the strawberry to appear at varying 2D positions within the image rather than always at the center. Domain randomization is applied over lighting conditions (intensity and direction) and strawberry plant geometry (size and shape), with a randomized seed for each scene to maximize dataset diversity. For each frame, we collect the RGB image, 2D bounding box, 3D bounding box (which provides $\mathbf{T}_{local \rightarrow world}$), and camera extrinsic (which is $\mathbf{T}_{world \rightarrow cam}$), all provided directly by the simulator without manual annotation. The 6D pose ground truth is then derived following the same formulation as the real-world dataset (Eq. (2)). Licensing rights to the strawberry plant model are held by one of the authors and do not permit open redistribution, so the model and the rendered synthetic dataset are available upon request.

B. Baseline Architecture

The overall architecture is shown in Fig. 3. We evaluate three backbone encoders representing different architectural and pretraining settings: ResNet-101 [17] as a CNN baseline, ViT-B/16 [18] as a supervised transformer baseline, and DINOv2-B [19] as a self-supervised vision foundation model.

All three backbones are evaluated within a shared architecture, where spatial features extracted by the backbone encoder are fed into a DETR-style [20] transformer decoder with learned object queries. The decoder jointly predicts a 2D bounding box, rotation represented via the continuous 6D representation of Zhou et al. [21], and translation decoupled into in-plane (x, y) and depth (z) components.

Loss Function. Object queries are matched to ground-truth annotations via Hungarian matching with cost:

$$\mathcal{C} = \lambda_{\text{bbox}} \mathcal{C}_{\text{bbox}} + \lambda_{\text{giou}} \mathcal{C}_{\text{giou}} \quad (4)$$

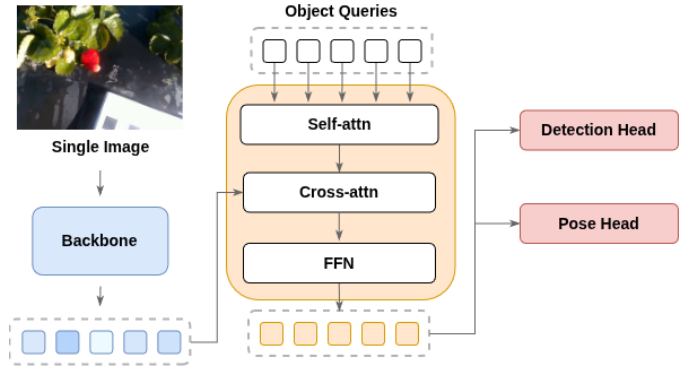


Fig. 3. Overview of the baseline architecture. Given a single RGB image, a backbone encoder extracts patch-level features, which are then processed by a transformer decoder with learnable object queries to jointly predict 2D detections and 6D poses via dedicated detection and pose heads.

where $\mathcal{C}_{\text{bbox}}$ is the L1 distance between predicted and ground-truth boxes and $\mathcal{C}_{\text{giou}}$ is the negative GIoU. Since only a single class is present, the classification term is omitted from the matching cost. Rotation and translation losses are computed only over matched pairs. The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \alpha(t)(\lambda_{\text{rot}} \mathcal{L}_{\text{rot}} + \lambda_{\text{xy}} \mathcal{L}_{\text{xy}} + \lambda_z \mathcal{L}_z) \quad (5)$$

where $\mathcal{L}_{\text{det}} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{bbox}} \mathcal{L}_{\text{bbox}} + \lambda_{\text{giou}} \mathcal{L}_{\text{giou}}$, $\mathcal{L}_{\text{cls}}$ is the object-vs-background classification loss, $\mathcal{L}_{\text{rot}}$ is the geodesic loss on rotation matrices, and $\mathcal{L}_{\text{xy}}$, $\mathcal{L}_z$ are SmoothL1 losses on in-plane and depth translation, respectively.

Since object queries must first learn to reliably localize strawberries before pose predictions become meaningful, pose losses are activated gradually via a warmup factor $\alpha(t) = \min(t/T_{\text{warmup}}, 1)$, where t is the current epoch and T_{warmup} is the warmup duration.

IV. EXPERIMENTS

A. Experimental Setup

Hardware. All models are trained using a single NVIDIA B200 GPU on the HiPerGator computing cluster, the University of Florida’s high-performance computing system.

Dataset Split. The real-world dataset is partitioned at the frame (image) level into training (10,040 images), validation (1,000 images), and test (1,000 images) sets. Validation and test sets consist exclusively of real-world images, and their indices are fixed prior to all experiments. The training set size is fixed at 10,040 images, with the proportion of real-world data varied from 0% (synthetic-only) to 100% (real-only), with the remainder drawn from the synthetic dataset. We denote the mixed settings as $R1$ – $R9$, corresponding to 10%–90% real-world training data, respectively.

Training Details. All models are trained using the AdamW optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . A StepLR scheduler reduces the learning rate every 30 epochs, and gradient clipping is applied with a maximum norm of 0.1. All backbones train for 50 epochs, using a batch size of 256. DINOv2-B is kept frozen, so only the decoder

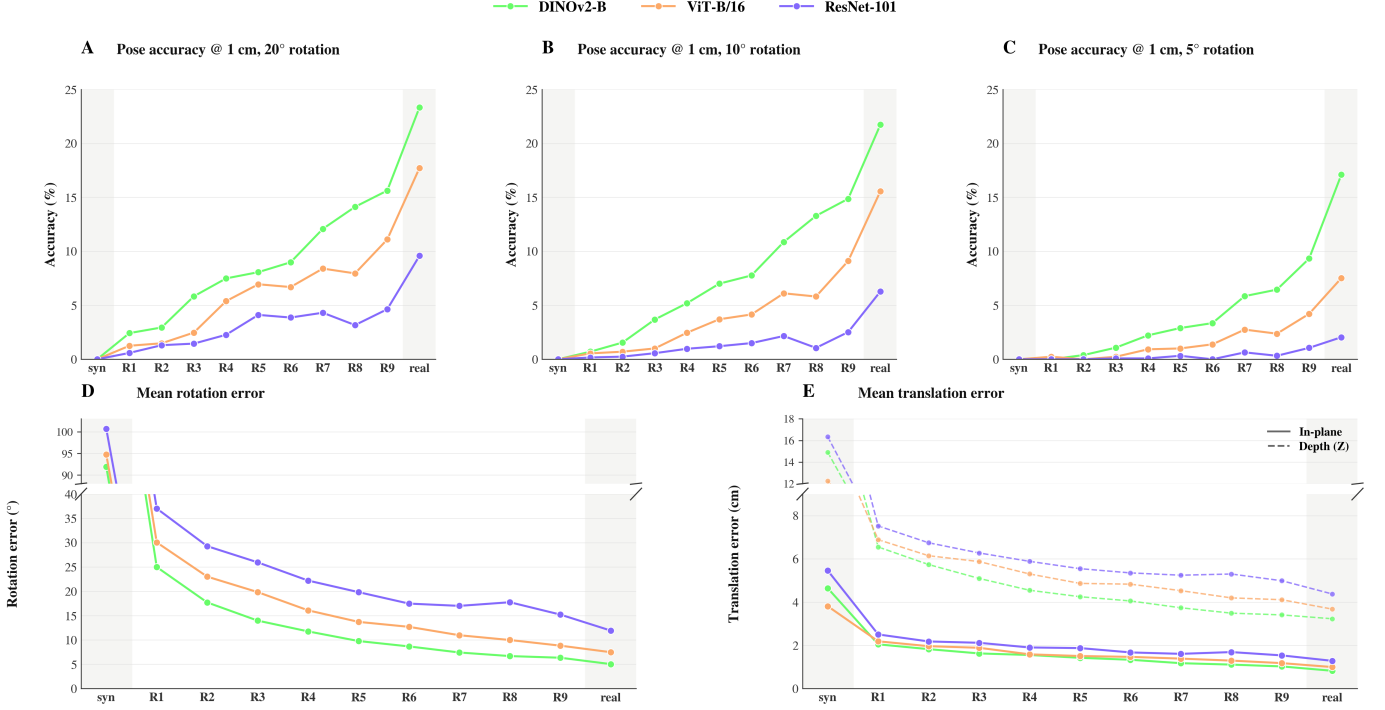


Fig. 4. Baseline results across backbone architectures and training configurations. *Syn* denotes synthetic-only training; *R1*–*R9* denote mixed training sets with 10%–90% real-world data; *Real* denotes real-only training. Gray shaded regions indicate the synthetic-only and real-only training configurations. (A–C) Pose accuracy at the stricter, grasp-relevant $\tau_t = 1$ cm translation threshold, with rotation thresholds of 20° , 10° , and 5° , respectively. (D) Mean rotation error. (E) Mean translation error decomposed into in-plane ($\sqrt{\epsilon_x^2 + \epsilon_y^2}$) and depth (ϵ_z) components. Panels D and E use a broken y -axis, since the synthetic-only value lies far outside the range of the real-data mixtures.

parameters are optimized, whereas ResNet-101 and ViT-B/16 are fine-tuned with the backbone updated at $0.1 \times$ the base learning rate to preserve pretrained representations. This asymmetry should be kept in mind when comparing backbones: DINOv2-B attains the strongest results in our experiments despite receiving no backbone adaptation at all. To stabilize early training, pose losses are activated via a linear warmup with $T_{\text{warmup}} = 20$ epochs, i.e., $\alpha(t) = \min(t/T_{\text{warmup}}, 1)$. **Loss Coefficients.** Following DETR [20], detection losses use coefficients of 1.0 (classification), 5.0 (bounding box L1), and 2.0 (GIoU).

Pose losses are weighted at 10.0 for rotation and in-plane translation, and 15.0 for depth translation, reflecting the inherently higher difficulty of monocular depth estimation.

B. Evaluation Metrics

Since strawberries are organic objects with substantial intra-class shape variation, a single mesh model cannot represent the geometry of individual real-world instances, making ADD (Average Distance of Model Points) and ADD-S (Average Distance of Model Points for Symmetric objects) unsuitable without instance-specific mesh models. We therefore adopt the following metrics.

Mean Rotation Error ($^\circ$) reports the average geodesic error between predicted and ground-truth rotations.

Stem-Axis Error ($^\circ$) reports the angle between the predicted and ground-truth stem-direction ($+z$) vectors, ignoring rota-

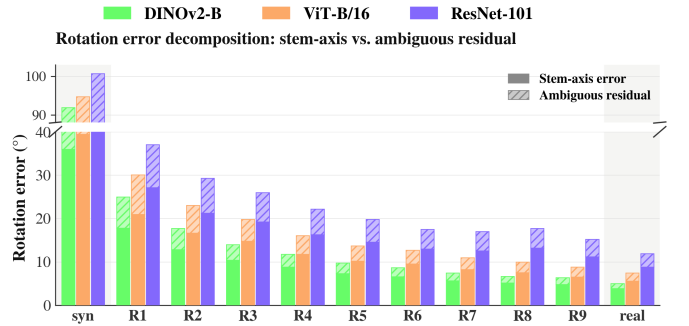


Fig. 5. Full rotation error versus the symmetry-aware stem-axis error (Sec. IV-B), which isolates the predicted stem-pointing direction about that axis. *Syn*, *R1*–*R9*, *Real*, and gray shaded regions as in Fig. 4. The plot uses a broken y -axis, as in Fig. 4D–E.

tion about that axis. Since the x/y azimuthal reference is not anatomically controlled (Sec. III-A) and strawberries are approximately rotationally symmetric about the stem, this symmetry-aware metric isolates the component of orientation (which way the stem points) that is both well-defined in the annotations and directly relevant to a grasp approach, complementing the full rotation error above. Our model regresses the full 6D rotation throughout; the stem direction reported here is one column of the predicted rotation matrix, so no axis-only detector is trained. Retaining the full rotation keeps

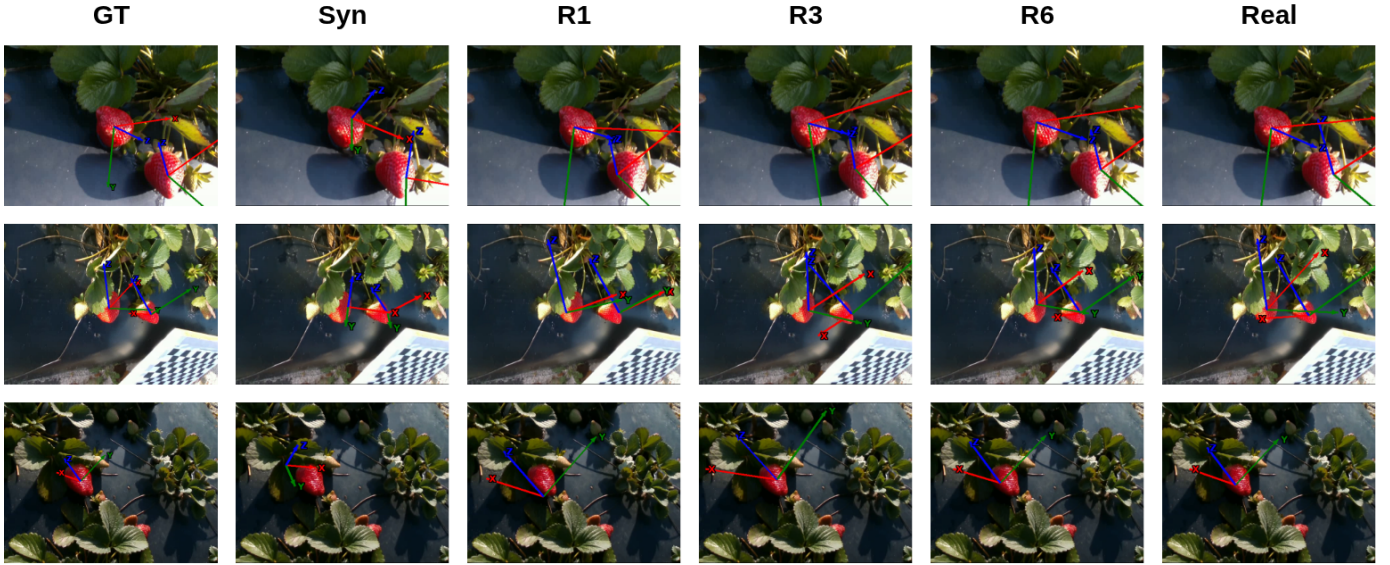


Fig. 6. Qualitative 6D pose estimation results across training configurations (DINOv2-B backbone). Columns correspond to: ground truth (GT), synthetic-only training (*Syn*), and models trained with increasing amounts of real data (*R1*, *R3*, *R6*, *Real*). Coordinate axes are color-coded: X (red), Y (green), Z (blue). *Syn*-only training yields poses far from GT across all scenes. Adding even a single real-data increment (*R1*) substantially reduces error, and predictions become progressively closer to GT as more real data is added.

the baseline comparable to standard 6D pose formulations and preserves the azimuthal component for grasp planners that may exploit it, for instance to steer a gripper around the calyx. It also makes the two quantities separable, which we use in Sec. IV-C to examine how the rotation error divides between the stem direction and the azimuthal component across training configurations.

Mean Translation Error (cm) reports translation error in centimeters and is further decomposed into in-plane ($\sqrt{\epsilon_x^2 + \epsilon_y^2}$) and depth components (ϵ_z).

Pose Accuracy is the fraction of predictions for which both the translation error and rotation error fall below their respective thresholds τ_t and τ_r . As a concrete point of reference, a ripe commercial strawberry is typically about 3–4 cm in diameter; a 1 cm error therefore corresponds to roughly a quarter to a third of the fruit’s diameter, large enough to plausibly affect grasp alignment, motivating our choice of 1 cm as a deliberately strict, task-scale-grounded threshold rather than an arbitrary one. We evaluate translation accuracy at this stricter, grasp-relevant $\tau_t = 1$ cm threshold throughout, chosen to make the gap between estimator capability and task requirement explicit rather than leaving it implicit. We vary $\tau_r \in \{5^\circ, 10^\circ, 20^\circ\}$ to assess rotational accuracy at multiple strictness levels.

C. Results

Synthetic-only training yields poor real-world transfer.

Models trained exclusively on synthetic data achieve 0.0% pose accuracy on real agricultural field images across all evaluated backbones and thresholds. Mean rotation errors exceed 90° for all backbones (DINOv2-B: 91.90° , ViT-B/16: 94.72° , ResNet-101: 100.68°), consistent with the synthetic-only values in Fig. 4D. This demonstrates that even synthetic

data with scene-level realism is insufficient for strawberry 6D pose estimation on an actual farm.

A small fraction of real data yields most of the improvement. Introducing just 10% real data (*R1*) causes a dramatic performance jump across all backbones. DINOv2-B’s mean rotation error drops from 91.90° to 25.00° , a 73% reduction from a single increment (Fig. 4D). Performance continues to improve as more real data is added, with gains largest at *R1* and diminishing progressively thereafter; no single saturation point is evident across all backbones.

Depth error is the dominant translation bottleneck. Depth translation error consistently exceeds in-plane error across all conditions, reflecting the fundamental depth ambiguity of monocular RGB-only 6D pose estimation. Under real-only training, DINOv2-B achieves an in-plane error of 0.83 cm but a depth error of 3.23 cm, roughly $4\times$ larger (Fig. 4E) and already spanning roughly 80% of the fruit’s width. Notably, in-plane error is already manageable at *R1* (~ 2 cm) and changes little thereafter, whereas depth error decreases more gradually and remains the primary bottleneck throughout. Decomposing the accuracy criterion confirms which term binds. Under real-only training, 92.5% of DINOv2-B’s predictions fall within the 10° rotation threshold, but only 23.3% fall within the 1 cm translation threshold, and the joint accuracy of 21.7% is nearly identical to the latter; the same ordering holds for ViT-B/16 (81.7% versus 17.7%, joint 15.6%) and ResNet-101 (52.8% versus 10.8%, joint 6.3%). Translation, and depth within it, is therefore what limits pose accuracy at this threshold, while rotation is rarely the failing criterion. Closing this gap likely requires additional depth cues (stereo, RGB-D, or multi-view geometry) beyond monocular RGB, which we leave for future work.

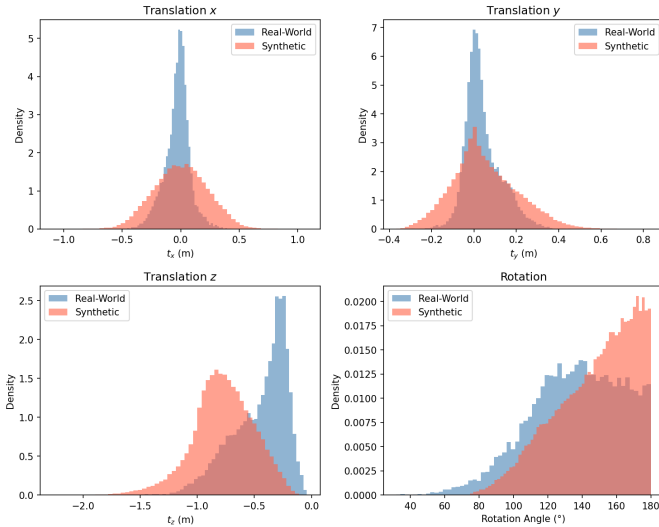


Fig. 7. Pose distributions of the real-world and synthetic datasets, including in-plane translations (t_x , t_y), depth translation (t_z), and rotation angle, defined as the geodesic magnitude of $\mathbf{R}_{local \rightarrow cam}$. The distributions reveal domain shifts in translation spread, depth range, and orientation coverage.

The azimuthal share of rotation error depends strongly on training configuration. The stem-axis error is consistently lower than the full rotation error across every backbone and training configuration (Fig. 5), but the size of the gap differs sharply between conditions. Under real-only training it is small in absolute terms: DINOv2-B’s full rotation error of 5.04° falls to 3.87° when azimuthal rotation about the stem is excluded, a difference of 1.17° (23%), with ViT-B/16 and ResNet-101 showing comparable relative reductions of 26% (1.93° and 3.10°). Under synthetic-only training the gap is far larger: DINOv2-B’s full rotation error of 91.90° falls to 35.98° (61% lower), with ViT-B/16 showing a similarly large 58% reduction and ResNet-101 a 52% reduction. Under real-only training the two metrics nearly coincide, so the rotation accuracy reported above is not an artifact of azimuthal ambiguity. The large azimuthal share under synthetic-only training should be read with care: at rotation errors near 90° the orientation is poorly recovered in any case, and a large azimuthal component is expected simply as a consequence.

Qualitative Analysis. Fig. 6 visualizes predicted poses across training configurations (DINOv2-B backbone). Synthetic-only predictions exhibit large axis misalignment across all scenes, while progressive convergence toward GT is visually apparent as real data increases.

D. Dataset Analysis

Ground-Truth Accuracy. The accuracy of the real-world ground-truth poses is affected by three quantifiable sources of error: camera calibration, PnP estimation, and COLMAP reconstruction, as well as manual 3D bounding box annotation, which we do not quantify. Camera calibration achieves an RMS reprojection error of 0.21 pixels. PnP estimation yields a median reprojection error of 0.50 pixels across 7,156 frames

(mean 1.17 pixels, right-skewed due to extreme viewpoints). COLMAP reports a mean reprojection error of 0.96 pixels across 99 valid sequences; 21 sequences were discarded due to reconstruction failure. Of the three quantified stages, COLMAP reconstruction shows the largest reprojection error; because annotation error is not quantified, we do not rank the sources overall.

Pose Distributions. Fig. 7 shows the pose distributions of both datasets. For in-plane translation (t_x , t_y), both distributions share a similar mean near zero, but differ substantially in spread: real-world instances exhibit a much narrower concentration, reflecting the constrained camera positioning during field collection, whereas synthetic instances are spread over a significantly wider range due to the unconstrained hemisphere sampling. The depth component t_z reveals a clear distributional shift between domains. Under the OpenGL convention adopted here (Sec. III-A) the camera looks along $-z$, so t_z is negative and we report distances as $|t_z|$. Real-world strawberries are captured at closer range, roughly 0.3 m from the camera, consistent with the actual robotic manipulation workspace, while synthetic instances are centered farther away at roughly 0.7 m, a consequence of sampling the synthetic camera distance uniformly over $[0.2, 1.0]$ m (Sec. III-A) rather than matching the distribution observed in the field. Rotation distributions also differ. We summarize each pose by the geodesic magnitude of $\mathbf{R}_{local \rightarrow cam}$, i.e., the angle of its axis-angle representation, so larger values indicate fruit orientations further from alignment with the camera frame. Synthetic data accumulates more heavily toward 160° – 180° than our field data, over-representing orientations far from alignment with the camera frame. These distributional mismatches in in-plane translation spread, mean depth, and rotation plausibly contribute to the sim-to-real gap quantified above, although we did not run a controlled experiment isolating their effect and therefore cannot attribute the gap to them.

V. CONCLUSION

We presented, to the best of our knowledge, the first real-world 6D pose ground truth dataset of red-stage strawberries collected at an actual strawberry farm (12,040 images), alongside a synthetic dataset rendered in NVIDIA Isaac Sim with scene-level realism and domain randomization. Baseline experiments across backbone encoders, under a monocular RGB setting, reveal that a substantial sim-to-real gap persists despite this improved simulation setup: synthetic-only training fails to transfer, and while a small amount of real data substantially improves rotation accuracy, depth error remains the limiting factor throughout. Closing this gap through simulation alone would require going beyond the scene-level realism and domain randomization explored here; real in-field data therefore remains necessary for both training and evaluating strawberry 6D pose estimation.

Several limitations remain: the dataset comes from a single farm and camera; annotation noise is uncharacterized, as each point cloud was labeled by one annotator; and the synthetic data would benefit from a more realistic strawberry plant

model and broader domain randomization over backgrounds, ground materials, and lighting.

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